

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.
8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.
9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.
10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.

EE1060: Differential Equations and Transform Techniques

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Topics covered :

1. Properties of Laplace Transform
2. Laplace Transform of some common signals

- **One-sided Laplace Transform:** For **causal signals** or **one-sided signals** or when dealing with **impulse response of a causal system**, the Laplace transform is defined as

$$\mathcal{L}\{cx(t)\} = \int_0^{\infty} x(t)e^{-st} dt \quad (1)$$

The Laplace transform defined using (1) is called the one-sided Laplace Transform.

- **Properties of Laplace Transform:**

- Scaling: If $\mathbf{X}(s)$ denotes the Laplace transform of $x(t)$, then the Laplace transform of $cx(t)$ is $c\mathbf{X}(s)$.

Proof:

$$\mathcal{L}\{cx(t)\} = \int_{-\infty}^{\infty} cx(t)e^{-st} dt = c \underbrace{\int_{-\infty}^{\infty} x(t)e^{-st} dt}_{\mathbf{X}(s)} = c\mathbf{X}(s) \quad (2)$$

This property holds for both two-sided and one-sided Laplace transforms. The Region of Convergence (ROC) is unaffected by the constant factor c , as it only scales the magnitude of the transform, not the region of convergence itself. Hence, the ROC for $cx(t)$ remains the same as the ROC for $x(t)$.

- Linearity: If $\mathbf{X}_1(s)$ and $\mathbf{X}_2(s)$ denote the Laplace transform of $x_1(t)$ and $x_2(t)$ respectively, then the Laplace transform of $ax_1(t) + bx_2(t)$ is $a\mathbf{X}_1(s) + b\mathbf{X}_2(s)$.

Proof:

$$\mathcal{L}\{ax_1(t) + bx_2(t)\} = \int_{-\infty}^{\infty} [ax_1(t) + bx_2(t)]e^{-st} dt = a \underbrace{\int_{-\infty}^{\infty} x_1(t)e^{-st} dt}_{\mathbf{X}_1(s)} + b \underbrace{\int_{-\infty}^{\infty} x_2(t)e^{-st} dt}_{\mathbf{X}_2(s)} = a\mathbf{X}_1(s) + b\mathbf{X}_2(s) \quad (3)$$

If $\mathbf{X}_1(s)$ and $\mathbf{X}_2(s)$ have their respective ROCs (say ROC_1 and ROC_2), then the ROC for the linear combination $a\mathbf{X}_1(s) + b\mathbf{X}_2(s)$ is the *intersection* of these two ROCs.

- Frequency shift: If $\mathbf{X}(s)$ denotes the Laplace transform of $x(t)$, then the Laplace transform of $e^{at}x(t)$ is $\mathbf{X}(s - a)$.

Proof:

$$\mathcal{L}\{e^{at}x(t)\} = \int_{-\infty}^{\infty} x(t)e^{at}e^{-st} dt = \int_{-\infty}^{\infty} x(t)e^{-(s-a)t} dt = \mathbf{X}(s - a) \quad (4)$$

For the two-sided Laplace transform, the ROC shifts accordingly. If the original ROC of $X(s)$ is ROC_X , then the ROC of $X(s-a)$ will be the same shape, but shifted in the s -plane by a . This means that the new ROC will be $\text{ROC}_X + a$, i.e., the region shifts by a along the real axis.

- Time shift: If $\mathbf{X}(s)$ denotes the Laplace transform of $x(t)$, then the Laplace transform of $x(t-\tau)$ is $e^{-s\tau}\mathbf{X}(s)$.

Proof:

$$\mathcal{L}\{x(t-\tau)\} = \int_{-\infty}^{\infty} x(t-\tau)e^{-st}dt = \int_{-\infty}^{\infty} x(t_1)e^{-s(t_1+\tau)}dt_1 = e^{-s\tau}\mathbf{X}(s) \quad (5)$$

The time-shifting operation $x(t-\tau)$ does not change the ROC.

- Scaling in time: If $\mathbf{X}(s)$ denotes the Laplace transform of $x(t)$, then the Laplace transform of $x(at)$, $a > 0$ is $\frac{1}{a}\mathbf{X}\left(\frac{s}{a}\right)$.

Proof:

$$\mathcal{L}\{x(at)\} = \int_{-\infty}^{\infty} x(at)e^{-st}dt = \frac{1}{a} \int_0^{-\infty} x(t_1)e^{-\left(\frac{s}{a}\right)t_1}dt_1 = \frac{1}{a}\mathbf{X}\left(\frac{s}{a}\right) \quad (6)$$

Scaling in time does not change the shape or nature of the poles, but it scales the s -variable by $\frac{1}{a}$, which means the ROC for $x(at)$ is the same as for $X(s)$, but scaled by $\frac{1}{a}$.

- Differentiation: If $\mathbf{X}(s)$ denotes the Laplace transform of $x(t)$, then the Laplace transform of $x'(t)$ is $s\mathbf{X}(s) - x(0)$.

Proof:

$$\begin{aligned} \mathcal{L}\{x'(t)\} &= \int_0^{\infty} x'(t)e^{-st}dt \\ &= [x(t)e^{-st}]_0^{\infty} - \int_0^{\infty} x(t)(-se^{-st})dt \\ &= [x(t)e^{-st}]_0^{\infty} + s \int_0^{\infty} x(t)e^{-st}dt \end{aligned} \quad (7)$$

If $x(t)$ is of **exponential order**, then

$$\begin{aligned} \mathcal{L}\{x'(t)\} &= \left(0 - x(0)e^{-s(0)}\right) + s\mathbf{X}(s) \\ &= s\mathbf{X}(s) - x(0). \end{aligned} \quad (8)$$

- Generalization for the n -th derivative: If $x(t)$ has the Laplace transform $\mathbf{X}(s)$, then the Laplace transform of the n -th derivative of $x(t)$, denoted by $x^{(n)}(t)$, is given by:

$$\mathcal{L}\{x^{(n)}(t)\} = s^n\mathbf{X}(s) - \sum_{k=0}^{n-1} s^{n-1-k}x^{(k)}(0). \quad (9)$$

- The Laplace transform of a signal is typically rational, meaning it can be expressed as a ratio of polynomials in

the complex variable s . Specifically, the Laplace transform $X(s)$ is represented as:

$$X(s) = \frac{N(s)}{D(s)} \quad (10)$$

where $N(s)$ is the numerator polynomial and $D(s)$ is the denominator polynomial. This rational form of the Laplace transform allows us to analyze the system's behavior by examining the locations of the roots of these polynomials. The key concepts are as follows:

- **Zeros of the Laplace Transform:** The roots of the numerator polynomial, where $N(s) = 0$, are called the **zeros** of $X(s)$. These are typically denoted by "o" in the complex plane. Zeros play an important role in determining the system's frequency response and overall behavior.
- **Poles of the Laplace Transform:** The roots of the denominator polynomial, where $D(s) = 0$, are referred to as the **poles** of $X(s)$. These are typically represented by "x" in the complex plane. Poles are crucial in determining the stability and dynamic behavior of the system.
- **Poles and Zeros at Infinity:** If the degree of the numerator polynomial $N(s)$ differs from that of the denominator polynomial $D(s)$, the system may exhibit either **poles at infinity** or **zeros at infinity**. This occurs under the following conditions:
 - * If the degree of the numerator is greater than the degree of the denominator, the Laplace transform has **zeros at infinity**.
 - * Conversely, if the degree of the denominator is greater than the degree of the numerator, the Laplace transform has **poles at infinity**.

These conditions reflect the behavior of the system at extreme frequencies and influence the system's stability and resonance characteristics.

In summary, the zeros and poles of the Laplace transform significantly influence the system's behavior in both the time and frequency domains. The locations of the zeros determine the system's frequency response, while the poles govern its stability and response to inputs. Understanding the distribution of these zeros and poles is fundamental in analyzing and designing control systems ¹.

¹refer to classroom scribble for examples on Laplace transform of some signals and stability analysis